ALP-ttbar Toy MC v1.7 — Preregistration: Three-Dimensional Two-Hypothesis Vertex Fit (Adding the r–z Opening Angle) for Displaced Light ALP → e^+e^-

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Zenodo: doi:10.5281/zenodo.23995668 · GitHub: github.com/wspaik5/ALP_ttbar_Toy_MC_v1.7_Preregistration_3D_Vertex_Fit
SHA-256 of the frozen README.md: `b538ea630b620aa881ca46f57320b530db439700e63c7ec0c50841e1d1ec3beb`

> **About this document.** This PDF reproduces the frozen preregistration text (README.md) unchanged and adds two figures made only from analytic formulas and truth-level quantities of the published v1.5 samples. No v1.7 simulation or fit had been run when it was written. **Disclaimer.** Independent personal toy study, not affiliated with the authors of arXiv:2610.38932 or arXiv:2306.08686, nor with the CMS, LHCb or ALICE Collaborations or the FCC study. Detector numbers are assumed; only qualitative trends and ratios are meaningful.

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Date frozen: 2026-11-06 (this file is committed to GitHub before any v1.7 simulation or fit is run)
Builds on: v1.6, https://doi.org/10.5281/zenodo.23952888 (preregistration https://doi.org/10.5281/zenodo.23942338); v1.5, https://doi.org/10.5281/zenodo.23924862; toy study motivated by arXiv:2610.38932
Zenodo (this preregistration, PDF): https://doi.org/10.5281/zenodo.23995668
GitHub: https://github.com/wspaik5/ALP_ttbar_Toy_MC_v1.7_Preregistration_3D_Vertex_Fit
Status: plan only. No v1.7 outcome exists at the time of this commit.
Amendment (2026-11-06, before any v1.7 simulation or fit): variant V-allpixel (Sec. 3.2) and prediction P7 added;
the first commit of this repository contained the same text without them.
PDF version: same text plus two figures drawn only from analytic formulas and truth-level numbers of the published v1.5 samples (no new simulation or fit).

0. Why v1.7

v1.5 and v1.6 used only the transverse (bending-plane) view. There, a free vertex absorbs a small ALP opening angle $\Delta\varphi$ by moving along the track, leaving a mismatch $\Delta\varphi^2/(2\Delta\omega)$ that is below the hit resolution at 17–50 MeV; and bremsstrahlung, which changes the curvature, makes about 10% of conversions look opening-angle-like. Both v1.5 and v1.6 listed the r–z view as the main untested ingredient (first item of the v1.6 future directions).

In the r–z view the situation is different:

- **The field does not bend the tracks** in r–z: $z = z_v + \cot\theta \cdot s$, a straight line in the transverse path length s.
- **A vertex shift cannot absorb a polar opening.** Two lines with different slopes $\cot\theta_1 \neq \cot\theta_2$ leave from one point; a fit that forces them to be parallel cannot reproduce them by moving the vertex. The opening $\Delta\cot\theta$ therefore enters the hits at first order, unlike $\Delta\varphi$ in the transverse view.
- **Bremsstrahlung barely changes the polar angle**: the radiated photon removes energy, not direction.
- **Photon conversions have $\Delta\cot\theta \approx 0$** (opening $\approx m_e/E$), apart from multiple scattering.

Truth-level facts used for planning (published v1.5 samples, base selection, no fit): the median polar opening $|\Delta\theta|$ of ALP pairs is 0.42 / 1.14 / 2.06 / 3.68 mrad at 17 / 50 / 100 / 200 MeV, and the fraction of pairs decaying at $r <$ 110 mm (at least two pixel layers with z measurement after the vertex) is 15.5% / 19.4% / 25.0% / 34.5%; for conversions it is 14.8%. In the default geometry only pixel layers measure z, so most of the r–z information is available only for vertices inside the pixel volume.

0a. Related work (prior art)

- **Using the r–z opening against conversions is standard.** Conversion reconstruction pairs tracks with a cut on $\Delta\cot\theta$, the opening in the r–z plane (e.g. the CMS conversion producer [B1]); low-mass dielectron analyses use the orientation of the opening relative to the field, φ_V [B2]. Three-dimensional vertex fitting is standard [B3].
- **Electron vs. muon displaced vertices** have been compared with a real detector (CMS, 8 TeV [B4]).
- **All-pixel trackers** are pursued by next-generation concepts such as ALICE 3 [B5] and the FCC-ee detectors [B6]; V-allpixel only mimics the z coverage of such trackers in this toy and makes no statement about their performance.

What v1.7 adds is limited to this series: it quantifies, in the preregistered toy of v1.0–v1.6, how much the r–z opening recovers the conversion-induced loss in reach for top-associated displaced ALP → e^+e^-. No claim of novelty for the method is made.

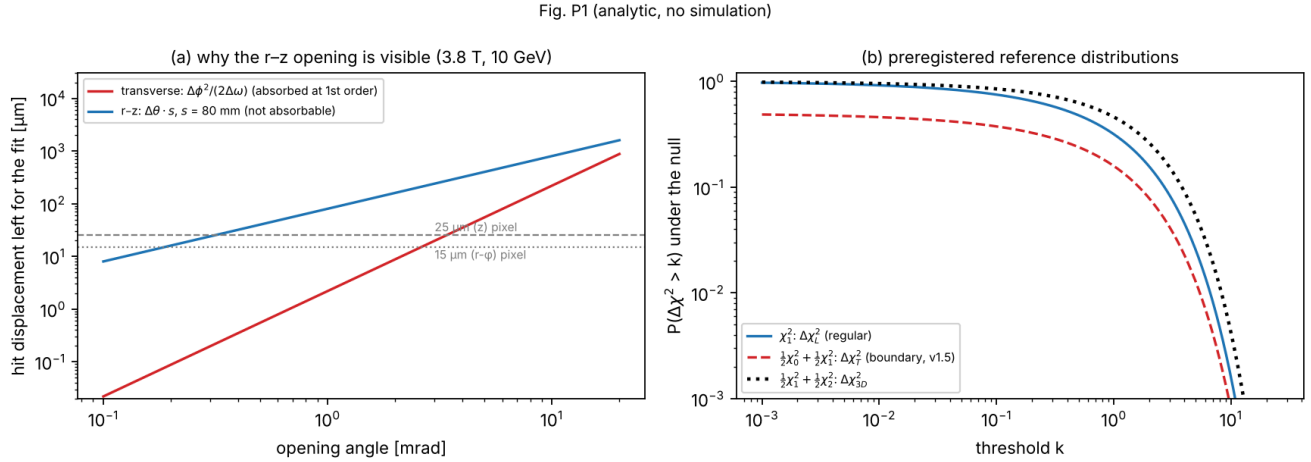


Fig. P1 (analytic, no simulation). (a) Hit displacement left for the fit by an opening angle, for two 10 GeV tracks at 3.8 T: in the transverse view a vertex shift absorbs Δφ at first order and leaves $\Delta\phi^2/(2\Delta\omega)$; in the r–z view a polar opening Δθ leaves Δθ·s (s = 80 mm shown), compared with the pixel resolutions. (b) The preregistered null reference distributions of Sec. 3.5.

1. Questions

- **Q1 (primary).** With the two-hypothesis selection extended to 3D, how large is the limit degradation from conversions at 17, 50, 100 and 200 MeV (cτ = 1 mm), compared with the transverse-only result on the same subsample (v1.6 config R: 127×, 68×, 28×, 8.2×)?
- **Q2.** How are $\Delta\chi^2$_L (r–z) and $\Delta\chi^2$_3D distributed for conversions and ALP pairs, and does bremsstrahlung leave the r–z test unaffected as expected?

2. Unchanged from v1.5/v1.6 (frozen)

- Transverse simulation and fits exactly as in v1.5/v1.6 config R (3.8 T, nominal material): the same samples, seeds, hits, H_conv (v1.4 fit (b)), H_ALP and $\Delta\chi^2$_T. The transverse part must reproduce v1.6 config R (Sec. 5).
- Statistics as in v1.6: first 100,000 base conversion pairs and first 60,000 base signal pairs per mass (whole chunks).
- Selection framework `scan_v15` (threshold k, veto width nσ, mass window), counting rule and metrics.

3. What is new (frozen)

3.1 Longitudinal simulation. Each track also gets z hits. The polar direction is propagated as a straight line in s; at every material surface after the vertex the polar angle receives an independent Highland scattering kick with the same θ_0 as the transverse kick; bremsstrahlung changes only the momentum (and thereby the transverse path, which is taken from the transverse simulation). The longitudinal random numbers come from a **separate generator** (seeds 1031 for conversions, 1041 for signal), so the transverse hits are bit-for-bit those of v1.5/v1.6.

3.2 z measurement.

- Default: the four pixel layers (r = 29, 68, 109, 160 mm) measure z with σ_z = 25 µm; strip layers do not measure z.
- Variant **V-stereo** (secondary): in addition, stereo strip layers TIB 1–2 (255, 339 mm) measure z with 0.25 mm and TOB 1–2 (608, 692 mm) with 0.5 mm.
- Variant **V-allpixel** (secondary): all 14 layers measure z with σ_z = 25 µm (and r–φ as in the default). The material, the field and the transverse simulation are unchanged. This is a toy stand-in for an all-pixel tracker of the kind pursued by next-generation concepts [B5, B6]; it changes only where z is measured, not the material budget of such trackers, which v1.6 showed to matter separately.

3.3 Longitudinal fit. For each hypothesis, after its transverse fit, the transverse path lengths s from the fitted vertex to each layer are computed along each fitted track. The r–z fit is then linear least squares:

- H_conv,L: z_v and one common slope c (tracks parallel in 3D);
- H_ALP,L: z_v, c_1, c_2. The covariance includes the z resolution and the scattering of the polar angle (lever arms along s, true momentum, as for the transverse term). $\Delta\chi^2_L = \chi^2_{conv,L} - \chi^2_{ALP,L}$ is defined when each track has at least two z hits; otherwise $\Delta\chi^2_L = 0$ (no r–z information).

3.4 Test statistic. $\Delta\chi^2_{3D} = \Delta\chi^2_T + \Delta\chi^2_L$.

3.5 Regularity check and reference distributions (v1.6 future direction (vi)).

- $\Delta\chi^2_L$: $\Delta\cot\theta$ enters linearly, the null value 0 is inside the parameter space, the fit is linear → regular; reference $\boldsymbol{\chi^2_1}$ (Wilks).
- $\Delta\chi^2_T$: as diagnosed in v1.5, boundary case → reference $\boldsymbol{\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1}$.

- Δχ²_3D: sum of the two independent parts → reference **½χ²₁ + ½χ²₂** (a Self–Liang-type mixture). Median 0.867, P(>3.84) = 9.8%, P(>9) = 0.69%. These references are used only for validation; the selection threshold k is scanned as before.

4. Selection (frozen)

`scan_v17` = `scan_v15` with Δχ²_3D in place of Δχ² (k ∈ {none, 1, 2, 4, 9, 16, 25, 49}). k = none reproduces the v1.6 config R degradation exactly. Reported side by side: Δχ²_T only (= v1.6 R), Δχ²_L only, Δχ²_3D; each with and without mass window (V-nomass); and the V-stereo and V-allpixel variants for Δχ²_3D.

5. Sanity gate (frozen)

Before any degradation is computed, for every sample:

1. **Identity.** Transverse vertex radius and Δχ²_T equal v1.6 config R within 10^{-6} for ≥ 99.9% of pairs.
2. **Fit success** ≥ 95% (transverse and longitudinal).
3. **Linear fit.** Fraction with Δχ²_L < -10^{-6} ≤ 0.1% (nested linear least squares cannot be negative).

If the gate fails, the code is fixed, the failure and the fix are reported, and the gate is re-run.

6. Validation (independent samples only; criteria frozen)

- **Z1 (null, no brems).** The v1.3 V4 collinear sample (seed 404) with z hits (longitudinal seed 1404), zero opening in 3D. Δχ²_L restricted to pairs with r–z information: median 0.30–0.65 and 3–8% above 3.84 (χ²₁). Δχ²_3D on the same pairs: median 0.60–1.15 and 6–14% above 3.84 (½χ²₁ + ½χ²₂).
- **Z2 (response, no brems).** Two 10 GeV tracks from a common vertex at r uniform in 30–100 mm, |η| < 1.0, Δφ = 0 and polar opening Δθ ∈ {0, 0.5, 1, 2} mrad (seed 808). The median Δχ²_L must increase monotonically with Δθ.
- **Z3 (propagation).** Without scattering, z hits agree with z_v + cotθ · s to better than 0.1 μm.

If a criterion fails, the code is fixed and the fix is reported.

Fig. P2 (truth-level numbers of the published v1.5 samples; no v1.7 simulation or fit)

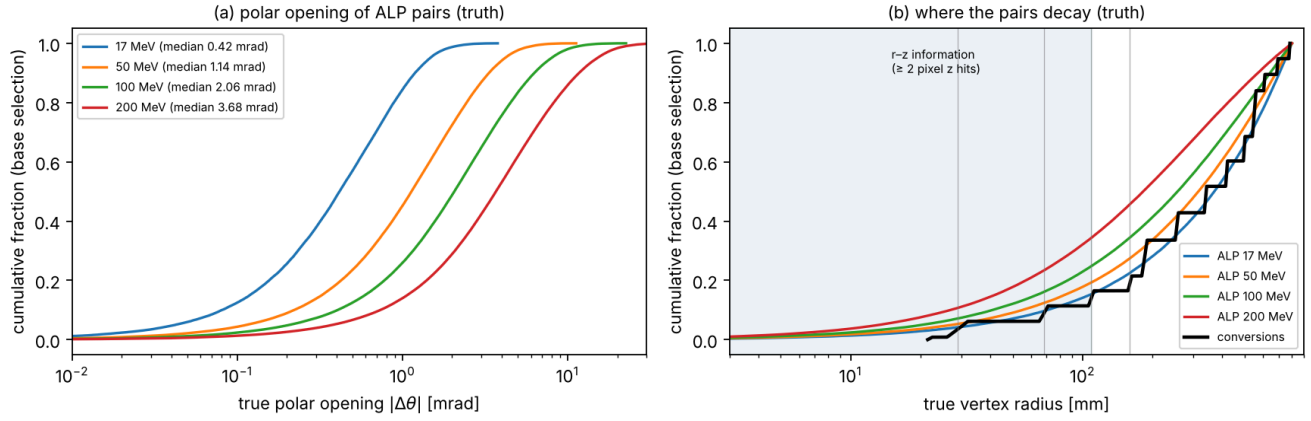


Fig. P2 (truth-level numbers of the published v1.5 samples; no v1.7 simulation or fit). (a) True polar opening of ALP pairs. (b) True vertex radius of ALP pairs and conversions (base selection); the shaded region has at least two pixel z measurements after the vertex.

7. Predictions (written before running)

Values are degradations with the Δχ²_3D selection, default z geometry, cτ = 1 mm, compared with v1.6 config R on the same subsample (127×, 68×, 28×, 8.2×).

- **P1 (17 MeV, primary).** Between **100× and 127×**: a gain of at most about 20%, because the median polar opening (0.42 mrad) is comparable to the r–z resolution and only about 15% of pairs decay inside the pixel volume.
- **P2 (200 MeV, primary).** Between **3× and 7×**.
- **P3 (100 MeV).** Between **12× and 25×**.
- **P4 (bremsstrahlung, mechanism).** For conversions (r < 800 mm) with r–z information, the fraction with Δχ²_L > 9 is **≤ 1%**, i.e. bremsstrahlung does not fake an r–z opening (v1.5 transverse: 9.6% with Δχ²_T > 9).
- **P5 (ALP, mechanism).** Among ALP pairs with r–z information, the fraction with Δχ²_L > 9 is **5–25%** at 17 MeV and **≥ 60%** at 200 MeV.
- **P6 (V-stereo, secondary).** Adding stereo z improves the 200 MeV degradation by **at least 10%** relative to the default geometry.
- **P7 (V-allpixel, secondary).** With z measured at every layer, at least 95% of pairs have r–z information, and the 17 MeV degradation is **at least 25% lower** than with the default geometry, but **still above 20×**: the r–z

opening becomes available for vertices outside the pixel volume, while multiple scattering (about 0.1–0.3 mrad per layer at 10 GeV, comparable to the 0.42 mrad median opening) and the conversion yield remain.

Each prediction is reported as confirmed or refuted against these statements. Refuted predictions are not re-framed.

8. Reporting

v1.7 note with a Figure 11: (a) $\Delta\chi^2_L$ distributions for conversions and ALP pairs with the χ^2_1 reference; (b) degradation versus mass for $\Delta\chi^2_T$ only, $\Delta\chi^2_L$ only, $\Delta\chi^2_3D$, V-stereo and V-allpixel, with the no-brems bound; (c) the fraction of pairs with r–z information and with $\Delta\chi^2_L > 9$ versus vertex radius. The note contains the Z1–Z3 validation, the prediction table, "Deviations from preregistration", "Limitations" and "Future directions" (Gaussian-sum filter on top of the 3D fit; dimuon control study; and, depending on the V-allpixel result, geometry studies such as a first layer closer to the beam and an all-pixel tracker with reduced material, combining the v1.6 material result with full z coverage).

9. Limitations known in advance

- All v1.0–v1.6 limitations apply (barrel only, idealised layers, no pattern recognition, toy material map, crude counting rule, selection optimised on the same samples, truth used for the material list and the scattering momentum, single-Gaussian bremsstrahlung model).
- The 3D fit is sequential (transverse first, then r–z with the transverse path lengths), not a joint 3D fit.
- The z resolution is a single number per layer type; cluster shapes, Lorentz drift and pixel-size effects are ignored.
- The pair mass is still generated independently of the hits (see V-nomass).
- Subsampled statistics; only trends and ratios are meaningful. No statement about real detectors is made.

[B1] CMS Collaboration, conversion producer (CMSSW `RecoEgamma/EgammaPhotonProducers`, parameter `DeltaCotTheta`); see also CMS Collaboration, Performance of photon reconstruction and identification with the CMS detector at √s = 8 TeV, JINST 10 (2015) P08010, arXiv:1502.02702. [B2] ALICE Collaboration, Dielectron production in proton-proton collisions at √s = 7 TeV, arXiv:1805.04391. [B3] R. Frühwirth, Application of Kalman filtering to track and vertex fitting, Nucl. Instrum. Meth. A 262 (1987) 444. [B4] CMS Collaboration, Search for long-lived particles that decay into final states containing two electrons or two muons in proton-proton collisions at √s = 8 TeV, Phys. Rev. D 91 (2015) 052012, arXiv:1411.6977. [B5] ALICE Collaboration, Letter of intent for ALICE 3, arXiv:2211.02491. [B6] N. Bacchetta et al., CLD – A Detector Concept for the FCC-ee, arXiv:1911.12230; The IDEA detector concept for FCC-ee, arXiv:2502.21223.